CLAUDE CERTIFIED ARCHITECT — FOUNDATIONS

Section 5: Model Context Protocol (MCP) Integration & Tool Design

Exam objectives covered

1. Distinguish between MCP resources and tools, and expose server content as resources to reduce exploratory tool calls and improve agent efficiency in cross-system queries.
2. Integrate MCP servers into Claude Code and agent applications by selecting the correct server scope, configuring authentication via environment variable expansion, and verifying tool discovery.
3. Write MCP tool descriptions that clearly distinguish each tool's purpose, input formats, use-case boundaries, and relationships to semantically similar tools, reducing misrouting and incorrect tool selection.
4. Configure the `tool_choice` parameter to guarantee tool invocation when structured output is required, and sequence multi-tool workflows so prerequisite data is obtained before dependent tools are called.

This section also covers broader MCP architecture concepts tested on the exam: client/server/host roles, transport mechanisms (stdio vs. HTTP/SSE), server scopes (local/project/user), authentication patterns, and tool naming/namespacing conventions.